

A Sequential Bayesian Algorithm for DOA Tracking in Time-Varying Environments*

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Abstract — This paper focuses on the Direction of arrival (DOA) tracking problem in dynamic environments where each source signal is modeled as a Gaussian process with time-varying mean and unknown covariance. In the presence of highly dynamic environment, benchmark algorithms usually have deteriorated performance. By treating the source signals as a function of the arrival angles, a sequential Bayesian tracking approach named Simultaneous angle-source update (SASU) is proposed based on the Maximum a posteriori (MAP) principle. The key feature of the proposed approach is to simultaneously update the arrival angles and the source signals in the Kalman filter step by converting the update process of the state vector into a joint optimization problem. An iterative Newton method to efficiently solve the joint optimization problem is proposed. The accuracy and robustness of the proposed SASU algorithm is demonstrated via simulations.

Key words — Time-varying, DOA tracking, Sequential Bayesian, Kalman filtering.

I. Introduction

Direction of arrival (DOA) tracking using sensor arrays is a classical problem in array signal processing due to its wide applications in radar, sonar and communications.

Two popular approaches for tracking arrival angles have been proposed in the past decades. The first approach is characterized by a two-step procedure that first estimates the arrival angles and then associates the estimated angles with the right targets. For DOA estimation, beamforming and signal-subspace techniques such as Multiple signal classification (MUSIC)^[1], Estimation of signal parameters via rotational invariance techniques (ESPRIT)^[2] as well as maximum likelihood principle based methods^[3,4] are widely used. Due to the need of multiple snapshots from different time instants, these approaches are not suitable for fast moving targets. Furthermore the extra steps of data association introduce unnecessary computational complexity and performance loss^[5]. The other approach is marked by applying tracking

algorithms, for example, Kalman filtering (KF), directly to arrival angles^[4]. This approach often requires more prior information, including proper statistical modeling, knowledge about the number of sources, and reasonable initial estimates of arrival angles. Since the measurement vector is a nonlinear function of the arrival angles, nonlinear Kalman filters, such as the Extended Kalman filter (EKF)^[6,7], Iterated extended Kalman filter (IEKF)^[8], Unscented Kalman filter (UKF)^[9] and Particle filter (PF)^[10] have been proposed.

In these approaches, two different signal models are considered according to different applications: the deterministic model and the stochastic model^[11]. The former assumes the tracked signals to be deterministic but unknown, and the latter characterizes the tracked signals as a stochastic process. In practical applications, some signals such as the acoustic signals from submarines and radar echoes from stealth objects are often assumed to be complex random process with time-varying mean and unknown covariance^[12,13]. In the presence of this time-varying source environment, the aforementioned benchmark algorithms usually fail or have deteriorated performance.

The main contribution of this paper is proposing a novel sequential Bayesian tracking approach named Simultaneous angle-source update (SASU), which focuses on the dynamic signal tracking problem discussed in Ref.[12] where each source signal is modeled as the mixture of a zero-mean Gaussian signal and a sinusoidal component with unknown coefficients and frequencies. For simplicity, we assume that the frequencies of the sinusoids and the covariance of the Gaussian noise have been estimated previously. Based on the Maximum a posteriori (MAP) principle in Bayesian statistics, the update process of the state vector is converted into a joint optimization problem at each discrete time instant during the tracking period. The key feature of our proposed approach is to simultaneously update the arrival angles and the source signals in the KF step by treating the source signals as a function of the arrival angles. With this treatment, the cost function of the joint optimization problem involves only the arrival an-

*Manuscript Received Mar. 2014; Accepted June 2014. This work is supported by the National Science Foundation for Distinguished Young Scholars (No.61025006) and the University Defence Research Center (UDRC), UK.

© 2015 Chinese Institute of Electronics. DOI:10.1049/cje.2015.01.023

gles. Subsequently, we design a iterative Newton method to efficiently minimize the cost function and update the state estimates. Simulation results show that our proposed SASU approach is more robust and accurate than the UKF, EKF and IEKF in the context of DOA tracking of highly dynamic sources.

The remainder of this paper is organized as follows. Section II introduces the system model and the state equation. The proposed method is detailed in Section III. Numerical simulations and conclusions appear in Section IV and Section V, respectively.

II. System Model and State Equation

Consider the DOA tracking problem of far-field and narrow-band source signals with a Uniform linear array (ULA) of m isotropic antennas. At the time instant t the measurement vector $\mathbf{y}(t) \in \mathbb{C}^m$ is given by

$$\mathbf{y}(t) = \mathbf{A}(\boldsymbol{\theta}(t)) \mathbf{s}(t) + \mathbf{w}(t) \quad (1)$$

where $\mathbf{s}(t) = [s_1(t), \dots, s_n(t)]^T$ and $\boldsymbol{\theta}(t) = [\theta_1(t), \dots, \theta_n(t)]^T$ denote n independent source signals and their corresponding arrival angles, respectively. The array direction matrix $\mathbf{A}(\boldsymbol{\theta}(t))$ and the direction vector $\mathbf{a}(\theta_n)$ are defined as $[\mathbf{a}(\theta_1(t)), \dots, \mathbf{a}(\theta_n(t))]$ and $[1, e^{-j2\pi(d/\lambda) \sin \theta_n} \dots, e^{-j2\pi(d/\lambda)(m-1) \sin \theta_n}]^T$, respectively. $\mathbf{w}(t) \in \mathbb{C}^m$ is the complex Gaussian noise vector.

Assume that each source signal is modeled as the mixture of a sinusoidal component $\boldsymbol{\psi}(\boldsymbol{\omega}, t) = [e^{j\omega_1 t}, e^{j\omega_2 t}, \dots, e^{j\omega_J t}]^T$ and a temporally zero-mean white Gaussian process $\boldsymbol{\zeta}(t) = [\zeta_1(t), \dots, \zeta_n(t)]^T$ with covariance $\boldsymbol{\Sigma}_{\boldsymbol{\zeta}}^{[12]}$, that is

$$\mathbf{s}(t) = \mathbf{C}\boldsymbol{\psi}(\boldsymbol{\omega}, t) + \boldsymbol{\zeta}(t) \quad (2)$$

where $\mathbf{C}$ is the $n \times J$ coefficient matrix of sinusoidal components for all sources and J is the total number of sinusoidal components with different frequencies in the received signals. Thus each source signal is a complex Gaussian process with time-varying mean, *i.e.*, $\mathbf{s}(t) \sim \mathcal{N}(\mathbf{C}\boldsymbol{\psi}(\boldsymbol{\omega}, t), \boldsymbol{\Sigma}_{\boldsymbol{\zeta}})$.

Further assume that each source consists of only one sinusoidal signal, with its frequency derived using the method in Ref.[12]. By some mathematical derivations, $\mathbf{s}(t)$ can be recursively expressed as

$$\mathbf{s}(t) = \mathbf{B}\mathbf{s}(t-1) + \mathbf{v}(t) \quad (3)$$

where $\mathbf{B} = \text{diag}(\exp(j\omega T_s)) \in \mathbb{C}^{n \times n}$ is a constant diagonal matrix and $\mathbf{v}(t)$ is a white Gaussian random process with covariance $2\boldsymbol{\Sigma}_{\boldsymbol{\zeta}}$.

Let $\dot{\theta}_i(t)$ be the angular speed for the i^{th} source signal, and $\dot{\boldsymbol{\theta}}(t) = [\dot{\theta}_1(t), \dots, \dot{\theta}_n(t)]^T$. For notational convenience, we use $\bar{\boldsymbol{\theta}}(t)$ to denote the vector $[\boldsymbol{\theta}^T(t), \dot{\boldsymbol{\theta}}^T(t)]^T$. The state equation can be written as^[14]

$$\bar{\boldsymbol{\theta}}(t) = \mathbf{F}\bar{\boldsymbol{\theta}}(t-1) + \mathbf{u}(t) \quad (4)$$

where $\mathbf{F} = \begin{bmatrix} \mathbf{I}_n & T_s \mathbf{I}_n \\ \mathbf{0} & \mathbf{I}_n \end{bmatrix}$, $\mathbf{I}_n$ is the $n \times n$ identity matrix and T_s is the sampling interval. $\mathbf{u}(t) \in \mathbb{R}^{2n}$ is assumed to

be a Gaussian random process with zero-mean and covariance matrix^[15,16]

$$\boldsymbol{\Sigma}_u = \sigma_u^2 \begin{bmatrix} \frac{T_s^3}{3} \mathbf{I}_n & \frac{T_s^2}{2} \mathbf{I}_n \\ \frac{T_s^2}{2} \mathbf{I}_n & T_s \mathbf{I}_n \end{bmatrix} \quad (5)$$

III. Proposed SASU Algorithm

1. Bayesian method for state update

Note that the signal vector $\mathbf{s}(t)$ and the measurement vector $\mathbf{y}(t)$ are complex vectors. It is convenient to rewrite the dynamic system in terms of real vectors. Define

$$\bar{\mathbf{B}} = \begin{bmatrix} \text{Re}(\mathbf{B}) & -\text{Im}(\mathbf{B}) \\ \text{Im}(\mathbf{B}) & \text{Re}(\mathbf{B}) \end{bmatrix}$$

$$\bar{\mathbf{A}}(\bar{\boldsymbol{\theta}}(t)) = \begin{bmatrix} \text{Re}(\mathbf{A}(\bar{\boldsymbol{\theta}}(t))) & -\text{Im}(\mathbf{A}(\bar{\boldsymbol{\theta}}(t))) \\ \text{Im}(\mathbf{A}(\bar{\boldsymbol{\theta}}(t))) & \text{Re}(\mathbf{A}(\bar{\boldsymbol{\theta}}(t))) \end{bmatrix}$$

Then one has

$$\begin{aligned} \bar{\mathbf{s}}(t) &= \bar{\mathbf{B}}\bar{\mathbf{s}}(t-1) + \bar{\mathbf{v}}(t) \\ \bar{\boldsymbol{\theta}}(t) &= \mathbf{F}\bar{\boldsymbol{\theta}}(t-1) + \mathbf{u}(t) \\ \bar{\mathbf{y}}(t) &= \bar{\mathbf{A}}(\bar{\boldsymbol{\theta}}(t))\bar{\mathbf{s}}(t) + \bar{\mathbf{w}}(t) \end{aligned} \quad (6)$$

where $\bar{\mathbf{s}}(t)$, $\bar{\mathbf{v}}(t)$, $\bar{\mathbf{y}}(t)$, $\bar{\mathbf{w}}(t)$ denote the real vectors consisting of their real part and imaginary part (*e.g.*, $\bar{\mathbf{s}}(t) = [\text{Re}(\mathbf{s}(t)) \text{Im}(\mathbf{s}(t))]^T$). We assume $\bar{\mathbf{v}}(t) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_v)$ and $\bar{\mathbf{w}}(t) \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_w)$ where $\boldsymbol{\Sigma}_v = \sigma_v^2 \mathbf{I}_{2n}$ and $\boldsymbol{\Sigma}_w = \sigma_w^2 \mathbf{I}_{2m}$. For simplicity, we assume that σ_v^2 and σ_w^2 have been estimated preliminary by the approach proposed in Ref.[12]. Note that the scalar notation of the subscript in the covariance matrices denotes that the covariance will not be updated with the time instant.

The aim of DOA tracking is to estimate $\hat{\boldsymbol{\theta}}(t)$ given $\bar{\mathbf{y}}(t)$ with predictions $\hat{\mathbf{s}}(t|t-1) = \bar{\mathbf{B}}\hat{\mathbf{s}}(t-1)$, $\hat{\boldsymbol{\theta}}(t|t-1) = \mathbf{F}\hat{\boldsymbol{\theta}}(t-1)$. Since the source signals vary with time, they are also taken inside the update process together with DOAs. Subsequently the state vector is updated according to the MAP principle, *i.e.*,

$$\left(\hat{\mathbf{s}}(t), \hat{\boldsymbol{\theta}}(t) \right) = \arg \max_{\bar{\mathbf{s}}, \bar{\boldsymbol{\theta}}} \Pr \left(\bar{\mathbf{s}}, \bar{\boldsymbol{\theta}} | \bar{\mathbf{y}}(t), \hat{\mathbf{s}}(t|t-1), \hat{\boldsymbol{\theta}}(t|t-1) \right).$$

To simplify the notations, let $\bar{\mathbf{s}}^0 = \hat{\mathbf{s}}(t|t-1)$, $\bar{\boldsymbol{\theta}}^0 = \hat{\boldsymbol{\theta}}(t|t-1)$ and $\bar{\mathbf{A}}(\bar{\boldsymbol{\theta}}) = \bar{\mathbf{A}}(\bar{\boldsymbol{\theta}}(t))$. For any vector $\mathbf{x}$ and a positive definite covariance matrix $\boldsymbol{\Sigma}$, define $\|\mathbf{x}\|_{\boldsymbol{\Sigma}^{-1}}^2 = \mathbf{x}^T \boldsymbol{\Sigma}^{-1} \mathbf{x}$. Then by the Gaussian assumption, one has

$$\begin{aligned} & -\log \Pr \left(\bar{\mathbf{s}}, \bar{\boldsymbol{\theta}} | \bar{\mathbf{y}}(t), \hat{\mathbf{s}}(t|t-1), \hat{\boldsymbol{\theta}}(t|t-1) \right) \\ &= c + \frac{1}{2} \|\bar{\mathbf{y}}(t) - \bar{\mathbf{A}}(\bar{\boldsymbol{\theta}}) \bar{\mathbf{s}}\|_{\boldsymbol{\Sigma}_w^{-1}}^2 \\ & \quad + \frac{1}{2} \|\bar{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}^0\|_{\boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}}^{-1}}^2 + \frac{1}{2} \|\bar{\mathbf{s}} - \bar{\mathbf{s}}^0\|_{\boldsymbol{\Sigma}_{\bar{\mathbf{s}}}^{-1}}^2 \end{aligned} \quad (7)$$

where c is a constant independent of $\bar{\boldsymbol{\theta}}$ and $\bar{\mathbf{s}}$, and $\boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t|t-1)}$ and $\boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t|t-1)}$ are the predicted estimate covariance matrices for $\bar{\boldsymbol{\theta}}(t)$ and $\bar{\mathbf{s}}(t)$ by their estimates at $t-1$ time instant respectively. Maximizing the posteriori probability is equivalent to minimizing Eq.(7).

This is a joint optimization problem with respect to $\bar{\mathbf{s}}$ and $\bar{\boldsymbol{\theta}}$. The key of SASU is to treat the source signals $\bar{\mathbf{s}}$ as a function of the arrival angles $\bar{\boldsymbol{\theta}}$. This is based on the observation that for a given $\bar{\boldsymbol{\theta}}$, the optimal $\bar{\mathbf{s}}$ to minimize Eq.(7) is unique:

take the partial derivative of Eq.(7) with respect to $\bar{s}$ and set it to zero, it can be verified that the optimal $\bar{s}$ is given by

$$\bar{s}(\bar{\theta}) = \bar{s}^0 + \left(\bar{A}^T(\bar{\theta}) \Sigma_w^{-1} \bar{A}(\bar{\theta}) + \Sigma_{\bar{s}(t|t-1)}^{-1} \right)^{-1} \cdot \bar{A}^T(\bar{\theta}) \Sigma_w^{-1} (\bar{y}(t) - \bar{A}(\bar{\theta}) \bar{s}^0) \quad (8)$$

where we use the notation $\bar{s}(\bar{\theta})$ to emphasize that the optimal $\bar{s}$ is determined by $\bar{\theta}$. As a result, the cost function in Eq.(7) can be written as

$$f(\bar{\theta}) = \|\bar{y}(t) - \bar{A}(\bar{\theta}) \bar{s}(\bar{\theta})\|_{\Sigma_w^{-1}}^2 + \|\bar{\theta} - \bar{\theta}^0\|_{\Sigma_{\bar{\theta}(t|t-1)}^{-1}}^2 + \|\bar{s}(\bar{\theta}) - \bar{s}^0\|_{\Sigma_{\bar{s}(t|t-1)}^{-1}}^2 \quad (9)$$

It is worth noting that the optimization is with respect to the arrival angles $\bar{\theta}$ only and that the source signals $\bar{s}$ will be automatically updated when $\bar{\theta}$ is updated.

2. Iterative Newton method for optimization

The focus of this subsection is to outline the efficient minimization process of the cost function in Eq.(9). To this end, we design a Newton method in which the second order derivative of f is not exactly computed. The basic idea is sketched as follows. At a given point, say $\bar{\theta}^k$, we approximate the cost function f by a quadratic function q . The optimal $\bar{\theta}$ that minimizes q admits a closed form, denoted by $\bar{\theta}^*$. It is clear that $\bar{\theta}^*$ may not be the minimizer of f . However, q is designed so that $\bar{\theta}^* - \bar{\theta}^k$ always gives a descent direction of f . Hence, one can search along the line $\bar{\theta}(\alpha) = \bar{\theta}^k + \alpha(\bar{\theta}^* - \bar{\theta}^k)$ to find an $\alpha \in [0, 1]$ such that $f(\bar{\theta}(\alpha)) < f(\bar{\theta}^k)$. Set $\bar{\theta}^{k+1} = \bar{\theta}(\alpha)$. Now one has an iterative procedure to minimize the cost function f . The quadratic approximation q is defined as follows. For a given point $\bar{\theta}^k$, define

$$\mathbf{G}^k = \nabla \left(\bar{A}(\bar{\theta}) \bar{s}(\bar{\theta}^k) \right) \quad (10)$$

where the gradient is with respect to $\bar{\theta}$. Note that in the approximation q , $\bar{s}$ is fixed to $\bar{s}(\bar{\theta}^k)$.

Define $\delta^k = \bar{\theta} - \bar{\theta}^k$ and $\bar{y}_r(t) = \bar{y}(t) - \bar{A}(\bar{\theta}^k) \bar{s}(\bar{\theta}^k)$. Then at a neighborhood of $\bar{\theta}^k$,

$$\begin{aligned} \bar{y}(t) - \bar{A}(\bar{\theta}) \bar{s}(\bar{\theta}) &\approx \bar{y}(t) - \bar{A}(\bar{\theta}^k) \bar{s}(\bar{\theta}^k) - \mathbf{G}^k (\bar{\theta} - \bar{\theta}^k) \\ &= \bar{y}_r(t) - \mathbf{G}^k \delta^k \end{aligned}$$

and the cost function in Eq.(9) can be approximated by

$$\begin{aligned} &\|\bar{y}_r(t) - \mathbf{G}^k \delta^k\|_{\Sigma_w^{-1}}^2 + \|\bar{\theta} - \bar{\theta}^0\|_{\Sigma_{\bar{\theta}(t|t-1)}^{-1}}^2 + c \\ &\approx \underbrace{\|\bar{y}_r(t) - \mathbf{G}^k \delta^k\|_{\Sigma_w^{-1}}^2 + \|\delta^k + \bar{\theta}^k - \bar{\theta}^0\|_{\Sigma_{\bar{\theta}(t|t-1)}^{-1}}^2}_{q} + c \end{aligned}$$

where $c = \|\bar{s}(\bar{\theta}^k) - \bar{s}^0\|_{\Sigma_{\bar{s}(t|t-1)}^{-1}}^2$ is a constant independent of $\bar{\theta}$. It is clear that q is a quadratic function of δ^k . When Σ_w^{-1} and $\Sigma_{\bar{\theta}(t|t-1)}$ are invertible, the approximation q admits a unique minimizer given by

$$\begin{aligned} \delta^* &= \left(\mathbf{G}^{kT} \Sigma_w^{-1} \mathbf{G}^k + \Sigma_{\bar{\theta}(t|t-1)}^{-1} \right)^{-1} \\ &\quad \cdot \left(\mathbf{G}^{kT} \Sigma_w^{-1} \bar{y}_r(t) - \Sigma_{\bar{\theta}(t|t-1)}^{-1} (\bar{\theta}^k - \bar{\theta}^0) \right) \quad (11) \end{aligned}$$

Besides the closed form for the minimizer, another important property of q is the following proposition.

Proposition Suppose that the vector δ^* in Eq.(11) is nonzero. Then there exists a strictly positive constant $\alpha > 0$ such that $f(\bar{\theta}^k + \alpha \delta^*) < f(\bar{\theta}^k)$.

Proof At each iteration the cost function f is approximated by the quadratic function q around the point $\bar{\theta}^k$. The vector δ^* points towards the minimum of the quadratic function and is also the minimizer of Eq.(9) at $\bar{\theta}^k$. In other words $\nabla f(\bar{\theta}^k) = -\delta^*$ or equivalently $\langle \delta^*, -\nabla f \rangle > 0$. According to the first order Taylor series expansion centered in $\bar{\theta}^k$ we obtain

$$f(\bar{\theta}^k + \alpha \delta^*) = f(\bar{\theta}^k) + \alpha (\nabla f|_{\bar{\theta}^k})^T \delta^* + o(\alpha \delta^*) \quad (12)$$

For a given $\bar{\theta}^k$, $\|\bar{s}(\bar{\theta}^k) - \bar{s}^0\|_{\Sigma_{\bar{s}(t|t-1)}^{-1}}$ is a constant independent of $\bar{\theta}$ and by expanding Eq.(9) we obtain

$$\begin{aligned} f(\bar{\theta}^k) &= [\bar{y}(t) - \bar{A}(\bar{\theta}^k) \bar{s}(\bar{\theta}^k)]^T \Sigma_w^{-1} [\bar{y}(t) - \bar{A}(\bar{\theta}^k) \bar{s}(\bar{\theta}^k)] \\ &\quad + [\bar{\theta}^k - \bar{\theta}^0]^T \Sigma_{\bar{\theta}(t|t-1)}^{-1} [\bar{\theta}^k - \bar{\theta}^0] \end{aligned}$$

Since $\partial \bar{y}(t) / \partial \bar{\theta}^k = 0$ and from Eq.(10) we obtain

$$\begin{aligned} \nabla f|_{\bar{\theta}^k} &= -\mathbf{G}^{kT} \Sigma_w^{-1} (\bar{y}(t) - \bar{A}(\bar{\theta}^k) \bar{s}(\bar{\theta}^k)) \\ &\quad + \Sigma_{\bar{\theta}(t|t-1)}^{-1} (\bar{\theta}^k - \bar{\theta}^0) \end{aligned}$$

For simplicity, let

$$\begin{aligned} \mathbf{a} &= \mathbf{G}^{kT} \Sigma_w^{-1} \bar{y}_r(t) \in \mathbb{R}^{2n \times 1} \\ \mathbf{b} &= \Sigma_{\bar{\theta}(t|t-1)}^{-1} (\bar{\theta}^k - \bar{\theta}^0) \in \mathbb{R}^{2n \times 1} \\ \mathbf{c} &= \mathbf{G}^{kT} \Sigma_w^{-1} \mathbf{G}^k + \Sigma_{\bar{\theta}(t|t-1)}^{-1} \in \mathbb{R}^{2n \times 2n} \end{aligned}$$

Consequently, one has

$$\begin{aligned} \nabla f|_{\bar{\theta}^k} &= -(\mathbf{a} - \mathbf{b}) \\ \delta^* &= \mathbf{c}^{-1}(\mathbf{a} - \mathbf{b}) \end{aligned}$$

and by Eq.(12)

$$\begin{aligned} f(\bar{\theta}^k + \alpha \delta^*) - f(\bar{\theta}^k) &= -\alpha(\mathbf{a} - \mathbf{b})^T \mathbf{c}^{-1}(\mathbf{a} - \mathbf{b}) \\ &\quad + o(\alpha \delta^*) \end{aligned}$$

For all α that are sufficiently small, it follows that $o(\alpha \delta^*) \ll |-\alpha(\mathbf{a} - \mathbf{b})^T \mathbf{c}^{-1}(\mathbf{a} - \mathbf{b})|$. That is to say, there exists a strictly positive constant α , such that

$$f(\bar{\theta}^k + \alpha \delta^*) < f(\bar{\theta}^k)$$

The significance of this property is that δ_k^* gives a descent direction of f . Hence, an iterative procedure to monotonically decrease f can be designed. The procedure used in this work is outlined in Algorithm 1.

Finally, we explicitly compute the gradient $\mathbf{G}^k$ defined in Eq.(10). Since $\bar{A}(\bar{\theta}) \bar{s}(\bar{\theta}^k) \in \mathbb{R}^{2m}$ and $\bar{\theta} \in \mathbb{R}^{2n}$, one has $\mathbf{G}^k \in \mathbb{R}^{2m \times 2n}$. From the definition of $\bar{A}(\bar{\theta})$, it can be verified that the $(i, \ell)^{th}$ entry of $\mathbf{G}^k$, denoted by $\mathbf{G}_{i, \ell}^k$, $1 \leq i \leq 2m$, $1 \leq \ell \leq 2n$, is given by

Algorithm 1 An Iterative Procedure to Minimize f

Input $\bar{\mathbf{y}}(t)$, $\bar{\boldsymbol{\theta}}^0 = \hat{\boldsymbol{\theta}}(t|t-1)$, $\bar{\mathbf{s}}^0 = \hat{\mathbf{s}}(t|t-1)$, $\boldsymbol{\Sigma}_w$, $\boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t|t-1)}$, error tolerance constants $\epsilon_1, \epsilon_2 \in \mathbb{R}^+$.

Output $\hat{\boldsymbol{\theta}}(t|t)$ and $\hat{\mathbf{s}}(t|t)$.

Iteratively perform the following: $k = 0, 1, \dots$

- 1) Compute δ^* according to Eq.(11).
- 2) Find $\bar{\boldsymbol{\theta}}^{k+1} = \bar{\boldsymbol{\theta}}^k + \alpha\delta^*$ by performing the following:
 - (a) If $\|\delta^*\|_2 < \epsilon_1$, set $\hat{\boldsymbol{\theta}}(t|t) = \bar{\boldsymbol{\theta}}^k$ and $\hat{\mathbf{s}}(t|t) = \bar{\mathbf{s}}(\bar{\boldsymbol{\theta}}^k)$, and quit this algorithm.
 - (b) Set $\alpha = 1$. If $f(\bar{\boldsymbol{\theta}}^k + \alpha\delta^*) \geq f(\bar{\boldsymbol{\theta}}^k)$, then decrease α via $\alpha = \alpha/2$ until $f(\bar{\boldsymbol{\theta}}^k + \alpha\delta^*) < f(\bar{\boldsymbol{\theta}}^k)$.
 - (c) Set $\bar{\boldsymbol{\theta}}^{k+1} = \bar{\boldsymbol{\theta}}^k + \alpha\delta^*$.
 - (d) If $|f(\bar{\boldsymbol{\theta}}^{k+1}) - f(\bar{\boldsymbol{\theta}}^k)| < \epsilon_2 f(\bar{\boldsymbol{\theta}}^k)$, then set $\hat{\boldsymbol{\theta}}(t|t) = \bar{\boldsymbol{\theta}}^{k+1}$ and $\hat{\mathbf{s}}(t|t) = \bar{\mathbf{s}}(\bar{\boldsymbol{\theta}}^{k+1})$, and quit this algorithm. Otherwise, start a new iteration.

$$\mathbf{G}_{i,\ell}^k = \frac{\partial (\bar{\mathbf{A}}(\bar{\boldsymbol{\theta}}) \bar{\mathbf{s}}(\bar{\boldsymbol{\theta}}^k))_i}{\partial \bar{\boldsymbol{\theta}}_\ell} = \begin{cases} 2\pi \frac{d}{\lambda} (i-1) \cos \theta_\ell [-\sin(2\pi \frac{d}{\lambda} (i-1) \sin \theta_\ell) \bar{s}_\ell \\ \quad + \cos(2\pi \frac{d}{\lambda} (i-1) \sin \theta_\ell) \bar{s}_{m+\ell}], & \text{if } i \leq m, \text{ and } \ell \leq n \\ 2\pi \frac{d}{\lambda} (i-1) \cos \theta_\ell [-\cos(2\pi \frac{d}{\lambda} (i-1) \sin \theta_\ell) \bar{s}_\ell \\ \quad - \sin(2\pi \frac{d}{\lambda} (i-1) \sin \theta_\ell) \bar{s}_{m+\ell}], & \text{if } i > m, \text{ and } \ell \leq n \\ 0, & \text{if } \ell > n \end{cases}$$

In Kalman filtering, one needs to update both the state and the covariance matrix of the state. That is, estimate $\boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t)}$ and $\boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t)}$ from $\boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t-1)}$ and $\boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t-1)}$. For the purpose of covariance update, we linearize the system at the point $\hat{\mathbf{s}}(t|t)$ and $\hat{\boldsymbol{\theta}}(t|t)$ (to simplify the notations, we use $\hat{\mathbf{s}}$ and $\hat{\boldsymbol{\theta}}$ henceforth). With respect to $\bar{\mathbf{s}}$, the system can be approximated by $\bar{\mathbf{y}}(t) = \bar{\mathbf{A}}(\hat{\boldsymbol{\theta}}) \bar{\mathbf{s}} + \bar{\mathbf{w}}$. Following from the covariance update step in the standard Kalman filter and Woodbury matrix inverse identity, one has

$$\begin{aligned} \boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t|t-1)}^{-1} &= \boldsymbol{\Sigma}_s^{-1} - \boldsymbol{\Sigma}_s^{-1} \left(\boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t-1)}^{-1} + \boldsymbol{\Sigma}_s^{-1} \right) \boldsymbol{\Sigma}_s^{-1} \\ \boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t)}^{-1} &= \boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t|t-1)}^{-1} + \bar{\mathbf{A}}^T(\hat{\boldsymbol{\theta}}) \boldsymbol{\Sigma}_w^{-1} \bar{\mathbf{A}}(\hat{\boldsymbol{\theta}}) \end{aligned}$$

where the matrix $\boldsymbol{\Sigma}_s$ is defined at the beginning of Section III. With respect to $\bar{\boldsymbol{\theta}}$, the system can be approximated by $\bar{\mathbf{y}}(t) = \bar{\mathbf{A}}(\hat{\boldsymbol{\theta}}) \hat{\mathbf{s}} + \mathbf{G}(\bar{\boldsymbol{\theta}} - \hat{\boldsymbol{\theta}}) + \bar{\mathbf{w}}$, where $\mathbf{G} = \nabla(\bar{\mathbf{A}}(\bar{\boldsymbol{\theta}}) \hat{\mathbf{s}})$ and the differentiation is performed at the point $\bar{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}$. Based on this linear approximation, one has

$$\begin{aligned} \boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t|t-1)}^{-1} &= \boldsymbol{\Sigma}_\theta^{-1} - \boldsymbol{\Sigma}_\theta^{-1} \mathbf{F} \\ &\quad \cdot \left(\boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t-1)}^{-1} + \mathbf{F}^T \boldsymbol{\Sigma}_\theta^{-1} \mathbf{F} \right)^{-1} \mathbf{F}^T \boldsymbol{\Sigma}_\theta^{-1} \\ \boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t)}^{-1} &= \boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t|t-1)}^{-1} + \mathbf{G}^T \boldsymbol{\Sigma}_w^{-1} \mathbf{G} \end{aligned}$$

where the constant matrix $\boldsymbol{\Sigma}_\theta$ is defined at the beginning of Section III. It is worth noting that instead of updating $\boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t)}$ and $\boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t)}$ directly, we choose to update $\boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t)}^{-1}$ and $\boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t)}^{-1}$. Note that $\boldsymbol{\Sigma}_{\bar{\mathbf{s}}(t)}^{-1}$ and $\boldsymbol{\Sigma}_{\bar{\boldsymbol{\theta}}(t)}^{-1}$ are in fact used in the objective function f . This update not only reduces computational complexity but also improves numerical stability according to our experience in simulations.

IV. Simulations

In our simulations, a ULA with 10 antennas is assumed to be placed along the positive x-axis and the first antenna is located at the origin of the coordinate system. The interval between two adjacent antennas is selected as $d = \lambda/2$. The simulated targets move parallel to the x-axis. The initial coordinates of the targets are $(-2000\text{m}, 2000\text{m})$, $(3000\text{m}, 2500\text{m})$, $(1500\text{m}, 3000\text{m})$ and the velocities are 500m/s , -300m/s and -500m/s respectively. Note that negative velocities are interpreted as moving in the opposite direction to the frame of reference. The total tracking period is 10 seconds while the sampling interval is 0.1 second. 100 random experiments were performed using a prior designed target movement. The coefficient matrix and the sinusoidal components are assumed as $\mathbf{C} = c_0 \cdot \mathbf{I}_3 = \sqrt{2}\mathbf{I}_3$ and $\boldsymbol{\psi}(\boldsymbol{\omega}, t) = [e^{-j0.4\pi t}, e^{j0.3\pi t}, e^{j0.5\pi t}]^T$ respectively, where $\mathbf{I}_3$ represents a 3×3 identity matrix. The Peak-signal-to-noise ratio (PSNR) of the source signal is defined as $10 \log(c_0^2/\sigma_v^2)$ in dB. At the beginning of the tracking process, the initial estimates of the DOAs and the angular speed vector are given by ESPRIT algorithm^[2] and the two-point interpolation method, respectively. Additionally the initial value of σ_w^2 and σ_v^2 are both assumed as one.

Fig.1 depicts the mean trajectories and Root mean square error (RMSE) profiles at PSNR = 0dB and $\sigma_w^2 = 0.2$ using our proposed algorithm. We can clearly see that the proposed SASU algorithm can accurately track each target trajectory at this level of PSNR. As expected, the RMSE tends to become larger when the DOAs of two targets approach gradually.

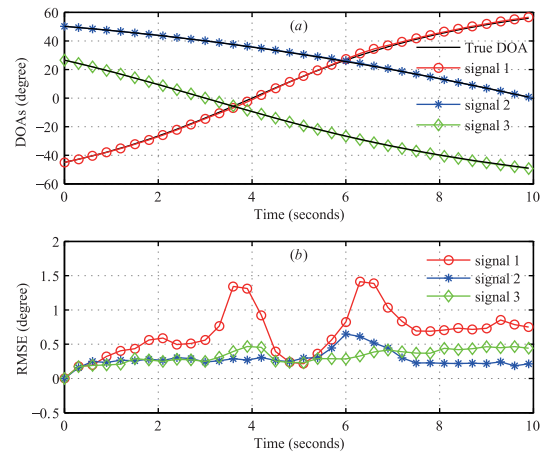


Fig. 1. Proposed SASU method tracking three targets. (a) Trajectories vs baselines; (b) RMSE

We also numerically compare the performance of SASU against three benchmark UKF, EKF and IEKF implementations at different levels of PSNR. Here we use averaged bias to evaluate the performance of each algorithm. The averaged bias is defined as

$$\text{bias} = \frac{1}{n \cdot T \cdot N} \sum_{i=1}^n \sum_{j=1}^T \sum_{k=1}^N \left| \hat{\boldsymbol{\theta}}_i(t) - \boldsymbol{\theta}_i(t) \right|$$

The number n , T and N indicate the source signal number, the sampling number during the whole tracking period and the amount of random experiments performed respectively. As

shown in Fig.2, the averaged deviations from the true angles of 100 simulations by EKF and UKF exceed 5° during the whole tracking period. This implies the intrinsic disability of these two methods in tracking the time-varying sources. In addition, it can be observed from the results that the proposed approach performs robustly even in a low signal-to-noise environment while IEKF is more sensitive to the signal noise.

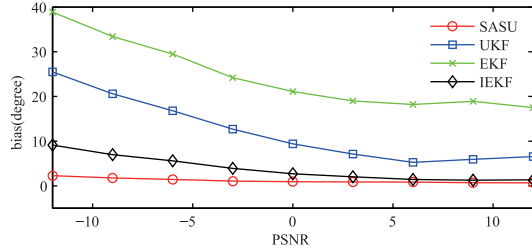


Fig. 2. Averaged bias comparison at different PSNR

To test the robustness of SASU, we also numerically compare the performance when σ_w^2 is fixed and σ_v^2 varies, and when σ_v^2 is fixed and σ_w^2 varies. Fig.3(a) and (b) illustrate the correct tracking ratio varying with σ_w^2 and σ_v^2 , respectively, where the correct tracking ratio is defined as the ratio of correct tracking against the number of experiments^[16]. A correct tracking is declared if the angle estimated is deviated from the true angle by less than 3° . It can be observed that both UKF and EKF are more sensitive to the signal noise and measurement noise. The results verify that SASU is much more robust than EKF and UKF especially when the uncertainty of the source signals is large.

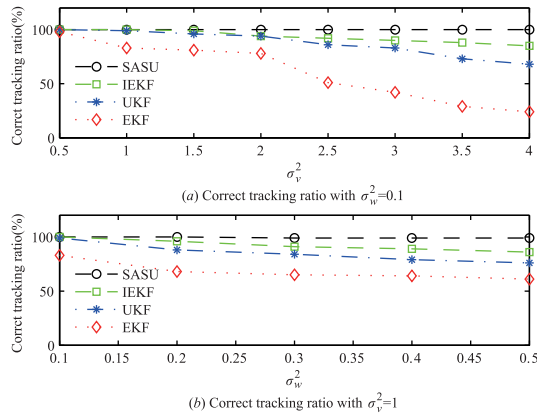


Fig. 3. Correct tracking ratio comparison. (a) σ_w^2 is fixed and σ_v^2 varies; (b) σ_v^2 is fixed and σ_w^2 varies

Finally, we would like to comment on the complexity of the proposed algorithm. The computational cost of the SASU algorithm is related to two factors. The first factor is related to the computation of the inverse matrix in each iteration. To avoid this problem, we use inverse of the covariance matrix as an iterative variable by the Woodbury equality as mentioned before. The other factor is the number of the iterations. Simulations show that ideal results can be achieved while the iterations run no more than 100 times.

V. Conclusion

We have presented a robust DOA tracking algorithm that

simultaneously updates the arrival angles and the source signals in the KF step and design a iterative Newton method to solve the joint optimization problem. Experimental results show the higher tracking accuracy and robustness of the proposed SASU approach when compared to benchmark UKF, EKF and IEKF algorithms in time-varying environments.

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